Department of Electronics Engineering

Walchand Institute of Technology, Solapur

A Data Analytics Report

(Couse Code: 23ECU6CC2T)

ON

DATATHON

Submitted by

1. **Pratik Bendsure**	**Roll No. 07**	**PRN No. 2303111015**
2. **Sidhant Bhure**	**Roll No. 09**	**PRN No. 2303111061**
3. **Saurabh Raut**	**Roll No. 54**	**PRN No. 2303111031**

Department of Electronics Engineering,
Walchand Institute of Technology, Solapur

(*An Autonomous Institute*)

Ashok Chowk, Solapur 413006

YEAR- 2025-26

Department of Electronics Engineering

Walchand Institute of Technology, Solapur

CERTIFICATE

This is to certify that the Mini Project entitled Datathon is a bonafide work carried out in partial fulfilment of the requirements of the Data Analytics course during the academic year 2025-26.

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2. Sidhant Bhure	**Roll No. 09**	**PRN No. 2303111061**
3. Saurabh Raut	**Roll No. 54**	**PRN No. 2303111031**

- **Class: T.Y B. Tech (Electronics and Computer Engineering)- Semester – VI**
- **Course Name: Data Analytics Course Code: *23ECU6CC2T***
- **Batch No: 01**

The report has been approved as it satisfactorily fulfils the academic requirements prescribed for the Mini Project course.

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Mr. R. P. Nagarkar
In-charge faculty

ABSTRACT

The insurance industry faces significant financial losses due to fraudulent claims, necessitating robust automated detection mechanisms. This project aims to develop a predictive analytics system capable of classifying car insurance claims as either legitimate or fraudulent. Utilizing a comprehensive dataset of historical car insurance claims, the data underwent rigorous preprocessing, including missing value imputation, label encoding, and feature selection to isolate the most predictive indicators of fraud.

We trained and evaluated advanced ensemble machine learning algorithms—specifically Random Forest and XGBoost—due to their robustness and ability to capture complex, non-linear tabular relationships. The experimental results demonstrated high discriminatory capability, with the XGBoost classifier outperforming Random Forest by achieving an accuracy of 81.00%, an F1-score of 0.5870, and an ROC-AUC of 0.8342.

To translate these predictive capabilities into actionable insights, a scalable, real-time interactive dashboard was developed using Streamlit. This tool allows end-users and insurance agents to input claim parameters dynamically, generating instant risk scores and visual breakdowns of exactly why a claim is flagged. Ultimately, this system provides a highly interpretable and efficient solution to allocate investigation resources accurately and expedite genuine claims.

1. INTRODUCTION AND PROBLEM DEFINITION

1.1 Background

Motor insurance fraud drains billions of dollars from the industry annually, which indirectly drives up premium costs for honest policyholders. Traditionally, insurance companies have relied on manual auditing and rigid, rule-based systems to detect false claims. This manual workflow is highly time-consuming, unscalable, and struggles to identify the increasingly sophisticated tactics used by modern fraudsters.To overcome these limitations, Machine Learning (ML) offers a highly effective, data-driven solution. By analyzing historical dataset patterns—such as incident details, vehicle types, and policyholder demographics—predictive models can instantly uncover hidden indicators of deceit. Implementing an AI-driven triage system allows insurance providers to automate the initial review process; this speeds up payouts for genuine claims to improve customer satisfaction, while accurately flagging suspicious claims for targeted human investigation.

1.2 Problem Statement

The insurance industry faces significant financial losses and operational inefficiencies due to fraudulent car insurance claims. Currently, the manual verification of these claims is slow, resource-intensive, and highly prone to human error.

1.3 Objectives

Develop a Predictive Model: To clean, preprocess, and analyze a historical car insurance dataset in order to train a robust classification model capable of distinguishing between legitimate and fraudulent claims.

Algorithm Evaluation & Comparison: To experiment with and evaluate multiple ensemble machine learning techniques (specifically Random Forest and XGBoost) to determine which algorithm yields the highest predictive performance (Accuracy, F1-score, and ROC-AUC).

Interactive Dashboard Development: To design and implement a responsive, completely functional web dashboard using Streamlit that enables users to input live claim metrics and receive a real-time likelihood estimate of fraud.

Ensure Model Explainability: To integrate interactive visual summaries and "Feature Importance" charts within the dashboard so that insurance agents can clearly understand why the AI flagged a specific claim.

Improve Operational Efficiency: To provide a proof-of-concept system demonstrating how predictive analytics can automate the triage process—saving investigation resources and preventing financial loss in the real world.

2. SYSTEM ARCHITECTURE AND METHODOLOGY

2.1 Module Description

Module 1: Data Acquisition & Preprocessing

This foundational module focuses on importing the raw Car Insurance Claims dataset and preparing it for machine learning consumption. It is responsible for handling missing data by imputing missing values (using the median for numerals and the mode for categorical data). It also performs feature reduction by discarding non-predictive identifiers (e.g., policy numbers, dates). Finally, this module utilizes Label Encoding techniques to translate textual categories into

a scaled numeric format required by decision-tree algorithms.

Module 2: Exploratory Data Analysis (EDA)
This module is dedicated to uncovering statistical patterns and underlying relationships within the dataset prior to model training. It involves generating correlation matrices to check for multicollinearity and visualizing data distributions. Specifically, the module investigates how distinct features, such as incident severity or auto deductibles, impact the mathematical likelihood of a claim being marked as fraudulent.

Module 3: Model Architecture & Training
This is the core computational module where predictive intelligence is generated. The preprocessed data is split into stratified training (80%) and testing (20%) boundaries. Two distinct ensemble models—**Random Forest Classifier** and **eXtreme Gradient Boosting (XGBoost)**—are initialized, trained, and fine-tuned. The module rigorously evaluates both algorithms against standard metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC). To finalize the pipeline, it serializes (saves) the trained models and their associated encoders into `.pkl` files for frontend use.

Module 4: Dashboard Integration & User Interface
This deployment module bridges the gap between raw AI code and the end-user. Utilizing Streamlit, it encapsulates the serialized models into a fully functional, premium dark-themed web application. The module manages the interactive frontend, allowing insurance agents to dynamically input live claim values through forms. It instantly calculates the fraud probability using the backend models and renders interactive Explanatory Visualizations (like Feature Importance charts) to clearly justify *why* the AI made its prediction.

2.2 Working Principle

1. **User Input:** An agent enters the details of an active claim (e.g., incident severity, policy details, vehicle damage) into the interactive Streamlit web dashboard.

2. **Data Transformation:** The backend instantly translates this text-based categorical input into a standardized numeric format using pre-trained label encoders.

3. **Algorithmic Inference:** This clean, numeric data is fed simultaneously into two active ensemble ML models (Random Forest and XGBoost) loaded in memory.

4. **Probability Prediction:** The pre-trained models route the input through hundreds of decision trees, calculating a final probability score (0% to 100%) that grades the likelihood of the claim being fraudulent.

5. **Visual Output:** The dashboard visually displays these risk scores instantly, alongside explainable data charts that highlight exactly which variables pushed the AI to its conclusion.

2.3 Algorithm Methodology

This project processes tabular data utilizing **Ensemble Machine Learning** techniques. The preprocessed dataset is mathematically split into an 80% training suite and a 20% unseen testing suite via stratified sampling. The system fundamentally explores two distinct multi-tree architectures:

1. **Random Forest (Bagging Strategy):** This approach systematically builds a randomized "forest" of 200 independent decision trees operating in parallel. By averaging the outputs of all trees upon inference, it significantly limits analytical variance and filters out dataset noise, delivering highly stable fraud baseline probabilities.
2. **XGBoost (Boosting Strategy):** Operating as a highly accelerated, sequential engine, XGBoost calculates 200 successive rounds of trees. Using complex gradient descent, each newly generated tree is algorithmically weighted to strictly correct the predictive errors made by its predecessors. This rapid self-correction allows XGBoost to learn extremely precise, non-linear fraudulent boundaries.

3. DATA PROCESSING AND ENGINEERING

3.1 Dataset Description

The dataset utilized for this project is the historical Car Insurance Claims Dataset (originally sourced from Kaggle). It contains exactly 1,000 recorded claim instances. The core objective of the model is to analyze the independent features below to predict the target variable: fraud_reported (Yes/No).
During preprocessing, strictly administrative strings (like policy numbers and exact dates) were dropped.

3.2 Preprocessing and Feature Engineering

1. **Handling Missing Data:** All ambiguous `'?'` characters were identified and converted to `NaN`. Missing numeric values were filled using the **median** (to prevent outlier skewing), while missing categorical values were filled using the **mode**.
2. **Feature Pruning:** Irrelevant identifier columns (e.g., `policy_number`, `incident_location`, and raw string dates) were permanently dropped to prevent the models from memorizing unnecessary administrative noise.
3. **Data Encoding:** The target variable (`fraud_reported`) was converted from 'Y/N' to a `1/0` binary. Finally, all remaining text-based columns (like `incident_type` and `severity`) were

processed through a **Label Encoder**. This transformation safely converted human-readable categories into the strict numerical format required by the decision-tree algorithms

4. SOFTWARE DESIGN AND IMPLEMENTATION

4.1 Tools Used

- Python 3.x
- Scikit-Learn, Pandas, NumPy
- XGBoost
- Streamlit
- Plotly, Matplotlib, Seaborn
- Joblib

4.2 Code Explanation and Integration

Model Training (train_models.py): Reads the historical CSV dataset, scrubs missing data, strategically maps categorical strings using Label Encoders, and trains both Random Forest and XGBoost classifiers. It evaluates algorithm performance and securely serializes the finalized models and encoders as .pkl artifacts.

Frontend & Integration (app.py): An integrated Streamlit application that generates the interactive web dashboard. It captures dynamic user inputs via input forms, pushes them securely through the pre-loaded .pkl models for real-time inference predictions, and natively renders the probability outputs as interactive Plotly visualizations without requiring an external backend API.

4.3 Testing Procedure

The backend API endpoints were tested using FastAPI's Swagger UI to ensure proper JSON parsing. The frontend was tested with edge cases to ensure stability.

5.3 Results and Analysis

The end-to-end integration was successfully deployed. Policy and incident metrics manually submitted by users on the Streamlit interface are accurately mathematically mapped and passed directly to the ensemble predictive models. The system seamlessly processes this data and returns highly actionable alerts ('**FRAUD DETECTED**' or '**LEGITIMATE CLAIM**') in real-time, completely unified alongside interactive probability risk scores and feature contribution charts.

5. CONCLUSION AND FUTURE SCOPE

5.1 Summary

The project successfully delivered a highly functional machine learning application capable of accurately predicting car insurance claim fraud, making robust ensemble models easily accessible to insurance agents via a premium, real-time visual dashboard.